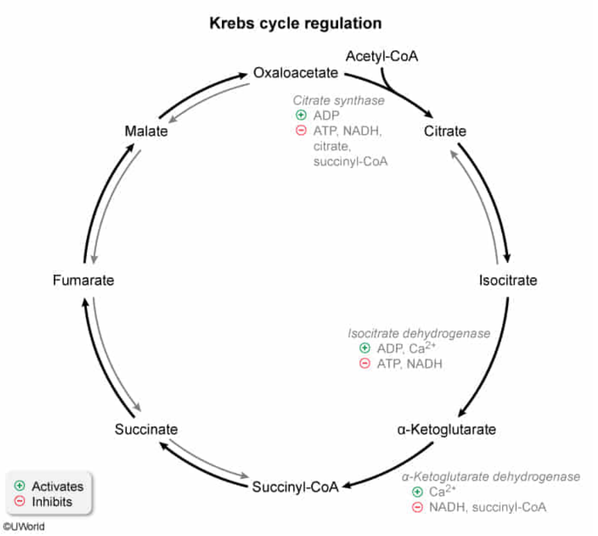
Which of the following molecules could decrease the activity of one or more Krebs cycle enzymes?

- A. Glucagon [15%]
- B. NADH [68%]
- C. Acetyl-CoA [10%]
- D. ADP [5%]

Correct
69% answered correctly

**Explanation:**




The **Krebs cycle**, also known as the citric acid cycle, **is tightly regulated** to produce ATP at the optimal rate for the needs of the cell. The cycle includes three **irreversible steps**, catalyzed by the enzymes citrate synthase, isocitrate dehydrogenase, and α-ketoglutarate dehydrogenase. The activity of each of these enzymes can be increased or decreased when specific molecules bind to the enzymes and alter their shape (allostery). Each of these molecules serves as an indicator of the energy state of the cell.

Molecules that are associated with **depleted energy** (eg, ADP) and muscle contractions (eg, calcium) indicate a need for additional ATP production. These molecules **allosterically activate** Krebs cycle enzymes to meet the demand for energy. Conversely, molecules that indicate that the Krebs cycle is producing **sufficient energy allosterically inhibit** the enzymes. These molecules include several products of the Krebs cycle and the electron transport chain such as ATP, citrate, succinyl-CoA, and NADH. **NADH decreases the activity** of all three of the regulated Krebs cycle enzymes.

**(Choice A)** Glucagon is a hormone that upregulates glucose production in the liver through glycogen breakdown (glycogenolysis) and gluconeogenesis. Although the liver does not use glucose as a source of acetyl-CoA for the Krebs cycle when glucagon is present, acetyl-CoA derived from fatty acid metabolism allows the cycle to remain active under these conditions.  The enzymes of the Krebs cycle are not under hormonal control.

**(Choice C)** Acetyl-CoA reacts with oxaloacetate to form citrate at the beginning of the Krebs cycle.  Its presence *increases* the overall rate of the cycle by increasing substrate concentration.

**(Choice D)** ADP is produced when ATP is hydrolyzed, and it signals to the cell that energy has been depleted.  It allosterically activates the Krebs cycle enzymes citrate synthase and isocitrate dehydrogenase, thereby *increasing* their activity.

**Educational objective:**
The Krebs cycle is allosterically regulated at the irreversible steps catalyzed by citrate synthase, isocitrate dehydrogenase, and α-ketoglutarate dehydrogenase.  The Krebs cycle enzymes are inhibited by NADH, ATP, citrate, and succinyl-CoA, and are activated by ADP and calcium.  These enzymes are not under hormonal control.

**Time spent:**0 sec
**Subject:**
Biochemistry

**QID**:401966
**System:**
Metabolic Reactions